

AP CALCULUS AB – UNIT 1

Limits and Continuity

AP-Style Review and Exam Practice



Name: _____ Class: _____ Date: _____

Directions

Section I Part A contains 16 non-calculator multiple-choice questions. Section I Part B contains 8 calculator-permitted multiple-choice questions. Section II contains four free-response questions. Show setup, exact reasoning, units, and justification.

Section I – Part A: No Calculator

1. Suppose $\lim_{x \rightarrow -1^-} f(x) = -4$ and $\lim_{x \rightarrow -1^+} f(x) = -2$. Which statement is correct?

- (A) $f(-1)$ must be undefined (B) $\lim_{x \rightarrow -1} f(x) = -2$
 (C) $\lim_{x \rightarrow -1} f(x) = -4$ (D) The two-sided limit does not exist

2. Evaluate $\lim_{x \rightarrow 3} \frac{(x-3)(x+1)}{x-3}$.

- (A) 0 (B) The limit does not exist
 (C) 4 (D) 3

3. Evaluate $\lim_{x \rightarrow 3} \frac{\sqrt{x} - \sqrt{3}}{x - 3}$.

- (A) The limit does not exist (B) 0
 (C) $2\sqrt{3}$ (D) $\frac{1}{2\sqrt{3}}$

4. If $|g(x)| \leq 2|x - 2|$ near $x = 2$, then $\lim_{x \rightarrow 2} g(x)$ equals

- (A) 0 (B) It cannot be determined
 (C) The limit does not exist (D) 2

5. For what value of k is $f(x) = \begin{cases} kx + 1, & x < 2 \\ x^2 + 4, & x \geq 2 \end{cases}$ continuous at $x = 2$?

- (A) $\frac{7}{2}$ (B) 8
 (C) $\frac{9}{2}$ (D) $\frac{5}{2}$

6. The function $f(x) = \frac{x+1}{(x-3)^2}$ has which behavior as $x \rightarrow 3$?

- (A) $f(x) \rightarrow 0$ (B) There is a removable discontinuity
 (C) The one-sided limits are opposite infinities (D) $f(x) \rightarrow +\infty$ from both sides

7. Find $\lim_{x \rightarrow \infty} \frac{5x^3 + 1}{2x^3 - 2}$.

- (A) $\frac{2}{5}$ (B) 0
 (C) $\frac{5}{2}$ (D) $+\infty$

8. A function f is continuous on $[0, 2]$, with $f(0) = -4$ and $f(2) = 5$. Which conclusion is guaranteed?

- (A) f is differentiable on $(0, 2)$ (B) There is exactly one such c
(C) f is increasing on $[0, 2]$ (D) There is at least one $c \in (0, 2)$ with $f(c) = -1$

9. Which discontinuity can be removed by redefining a single function value?

- (A) An oscillatory discontinuity (B) A hole where the finite two-sided limit exists
(C) A jump discontinuity (D) A vertical asymptote

10. If f is continuous at $x = 2$ and $\lim_{x \rightarrow 2} f(x) = 7$, then which must be true?

- (A) $f(2) = 7$ (B) The graph has a vertical asymptote
(C) $f(2)$ is undefined (D) $f'(2) = 7$

11. Suppose $\lim_{x \rightarrow 3^-} f(x) = 1$ and $\lim_{x \rightarrow 3^+} f(x) = 1$. Which statement is correct?

- (A) $\lim_{x \rightarrow 3} f(x) = 1$ (B) $\lim_{x \rightarrow 3} f(x) = 0$
(C) $f(3) = 1$ must hold (D) The limit does not exist

12. Evaluate $\lim_{x \rightarrow -3} \frac{(x - -3)(x + 5)}{x - -3}$.

- (A) The limit does not exist (B) 0
(C) 2 (D) -3

13. Evaluate $\lim_{x \rightarrow 1} \frac{\sqrt{x} - \sqrt{1}}{x - 1}$.

- (A) $\frac{1}{2\sqrt{1}}$ (B) 0
(C) $2\sqrt{1}$ (D) The limit does not exist

14. If $|g(x)| \leq 4|x - 2|$ near $x = 2$, then $\lim_{x \rightarrow 2} g(x)$ equals

- (A) 4 (B) The limit does not exist
(C) It cannot be determined (D) 0

15. For what value of k is $f(x) = \begin{cases} kx + 1, & x < 1 \\ x^2 + 4, & x \geq 1 \end{cases}$ continuous at $x = 1$?

- (A) 4 (B) 9
(C) 3 (D) 5

16. The function $f(x) = \frac{x + 1}{(x - 3)^2}$ has which behavior as $x \rightarrow 3$?

- (A) The one-sided limits are opposite infinities (B) $f(x) \rightarrow +\infty$ from both sides
(C) $f(x) \rightarrow 0$ (D) There is a removable discontinuity

Section I – Part B: Calculator Permitted

17. A calculator table near $x = 1$ gives $f(0.9000)=-3.3230$; $f(0.9900)=-3.4730$; $f(0.9990)=-3.4563$; $f(0.9999)=-3.3980$. Which is the best estimate of $\lim_{x \rightarrow 1^-} f(x)$?

- (A) The limit does not exist
(B) -3.323
(C) 1
(D) -3.398

18. Use a graphing calculator to estimate $\lim_{x \rightarrow 1.69} \frac{\sin(x - 1.69)}{x - 1.69}$.

- (A) 0
(B) The limit does not exist
(C) -1
(D) 1

19. A graph of $f(x) = \frac{x^2 - 16}{x - 4}$ shows a hole. What is the y-coordinate of the hole?

- (A) 8
(B) 0
(C) 9
(D) 4

20. Use numerical evidence to determine the sign of $\lim_{x \rightarrow 2^-} \frac{1}{x - 2}$.

- (A) $+\infty$
(B) $-\infty$
(C) The limit does not exist
(D) 0

21. A calculator table near $x = 1$ gives $f(0.9000)=3.7400$; $f(0.9900)=3.6900$; $f(0.9990)=3.6067$; $f(0.9999)=3.6650$. Which is the best estimate of $\lim_{x \rightarrow 1^-} f(x)$?

- (A) 3.665
(B) 3.74
(C) The limit does not exist
(D) 1

22. Use a graphing calculator to estimate $\lim_{x \rightarrow 2.93} \frac{\sin(x - 2.93)}{x - 2.93}$.

- (A) -1
(B) The limit does not exist
(C) 1
(D) 0

23. A graph of $f(x) = \frac{x^2 - 4}{x - 2}$ shows a hole. What is the y-coordinate of the hole?

- (A) 2
(B) 4
(C) 0
(D) 5

24. Use numerical evidence to determine the sign of $\lim_{x \rightarrow 1^-} \frac{1}{x - 1}$.

- (A) $-\infty$
(B) $+\infty$
(C) The limit does not exist
(D) 0

Section II – Free Response

Questions 1–2: Calculator permitted. Questions 3–4: No calculator.

FRQ1 – Limits from numerical data

A table gives $f(1.9) = 3.80$, $f(1.99) = 3.98$, $f(2.01) = 4.02$, and $f(2.1) = 4.20$, while $f(2) = 7$.

(a) Estimate the left- and right-hand limits and state whether the two-sided limit exists. [3 points]

(b) Explain why the value $f(2) = 7$ does not change the limit. [2 points]

(c) Redefine $f(2)$ so that the function is continuous at $x = 2$. [2 points]

FRQ2 – Algebraic limit and removable discontinuity

Let $g(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$.

(a) Evaluate $\lim_{x \rightarrow 3} g(x)$ and justify the algebraic step. [3 points]

(b) Define $g(3)$ so that the extended function is continuous. [1 points]

(c) Describe the graph before and after the redefinition. [2 points]

FRQ3 – Asymptotic behavior

Consider $h(x) = \frac{2x^2 + 5x}{x^2 - 4}$.

(a) Determine all vertical asymptotes and justify one-sided behavior. [4 points]

(b) Find the horizontal asymptote using a limit at infinity. [2 points]

(c) Explain the different meanings of the two types of asymptotes. [2 points]

FRQ4 – Intermediate Value Theorem

A continuous function p satisfies $p(1) = -3$, $p(4) = 5$, and $p(7) = -1$.

(a) Use IVT to guarantee at least two distinct zeros. [4 points]

(b) Can IVT prove that there are exactly two zeros? Explain. [2 points]

(c) State all hypotheses used. [2 points]